Paper 14: The Choice of Time Direction and the Stage for Amplitudes — the Marking Theorem, Derived Z ⊠ Phases, Inversion Holonomy, and the Canonical Counting Theorem

Author: Noriaki Kihara **Date**: June 12, 2026 **Version**: v1.1 (content and results unchanged from v1.0; nomenclature unified to Paper 0.5's displacement-record definition — ledger/register/bookkeeping mapped to conserved quantity / invariant / accounting) **DOI (this version)**: 10.5281/zenodo.20690059 / **Concept DOI**: 10.5281/zenodo.20665661 **License**: CC BY 4.0 **Series**: Dual Geometry of Wavelength Space and Frequency Space (Paper 14, sequel to Papers 1–13)

Abstract

Paper 13 showed that position and time can be read from the axiom inventory alone. Two questions remained — **why does a particular axis look like time**, and **where do amplitudes (complex numbers, phases) come from?** This paper theorematizes both within the inventory. Main results: (1) **Choosing a time direction = choosing a complex structure** (Theorem 1): choosing a time direction u inside the quaternions $\mathbb{H}$ selects the complex subfield $C_u \subset \mathbb{H}$, and the polar decomposition $q = |q|e^{u\theta}$ yields **the clock (modulus) and the phase (argument) simultaneously**. Complex numbers are not an axiom but the shadow of this choice. A proven companion identity (Proposition 1′ , with a three-line proof, range-unlimited): splitting the dressed vector $d_j = 2|k_j| + 1$ into the Gaussian integers $v_1 = (d_1 + id_2)/(1 + i)$, $v_2 = (d_3 + id_4)/(1 + i)$ gives $N(v_1) + N(v_2) = 2s$ with each $N(v_j)$ **odd** — the factor $(1 + i)$ joins the norm rung to the odd-Gaussian rung. (2) **The marking theorem** (Theorem 2): lattice-compatible quaternion structures number **exactly 16**, on which B ⊠ acts **transitively** with point stabilizers the 24-element ring-automorphism group ($384 = 16 \times 24$) — the real axis runs over all four axes. The principle "all axes are symmetric; t merely appears to have been selected by observation" becomes a theorem here. Of the first-order canonical readings defined per marking, the ones shared by all 16 are **the norm (R) and the central character ε (Q)** — completeness is asserted for that family of functions, not for arbitrary cell functions. (3) **Transport phases are derived** (Theorem 3): the coefficient ratio η between parent line and daughter cross line is fixed uniquely by the dictionary's trigonometric identities and **always lands on** $Z_4 = \{1, i, -1, -i\}$ (zero exceptions over all paths at s=9/11/13 — phase assignment rules became unnecessary). (4) **The identity of the forced bit** (Theorem 4): the structure of the channel-consistent aggregate demands exactly one bit of choice, and its identity is **not the arrow** (axiom 3; invariant under time reversal of comparisons) but the Z_2 **chain holonomy of canonical inversion** $g \to -g$ **= complex conjugation** — the system has two independent global Z_2's. The diagnostic quantity here is explicitly the **channel-consistent sum**, one particular aggregate, whose measure-theoretic status is adjudicated by Paper 16; the scope restriction is substantiated by a new check (at s=9 configuration granularity, W'_{531} is sector-independent, 576=576, while W'_{333} retains dependence, 40 vs 16 — the bit is not an artifact of the aggregate, but the index-2 formulation is stated at the channel-consistent level). (5) **The canonical counting theorem** (Theorem 5): how to count identical daughters is not a convention — configuration reading (state = set) forces **identical-particle symmetrization**, uniquely; the treatment of diagonal terms and decay branches does not affect the amplitude accounting (machine-verified 2×2 separation: ordering × diagonal). What standard theory installs as the symmetrization postulate appears here as a theorem of set structure — and the theorem is **granularity-robust** (the doubling acts at the level of per-configuration z_K). Appendix A collects the general

transport-algebra lemmas (single-axis branch-flip ±i lemma — 103,168 flips over three shells, zero exceptions; single-sign support; the charge-like parity theorem; the mirror-flip lemma). **This paper does not select an aggregation rule (measure) for path amplitudes** — it stops at the definition of η and path sets and their phase algebra (the measure line is Paper 16). No physical identification is made.

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1. Introduction

1.1 The questions

Gauge data held fixed (complete enumeration, same standard as Paper 13): the marking u, per-axis orientations Z_2^4, and the reference fragment (origin). All theorems below are statements about properties across gauge choices (transitivity; shared invariants).

At the endpoint of Paper 13 the four axes were fully symmetric (interchangeable, including R and Q) and t was a derived quantity. Then: why does some direction look like "time" to an observer? And since complex numbers exist nowhere among the axioms of Papers 5–12, while the record readout (branch channels, Paper 13, Theorem 3) handles complex coefficients — **where did i come from?** The answer: both are two faces of one and the same choice — **time direction = complex structure** — whose freedom, invariants, and the exactly-one-bit remainder can all be theorematized.

1.2 What is not claimed

(i) **This paper does not select an aggregation rule (measure) for path amplitudes.** It theorematizes the raw material (η , path sets, phase algebra); how they are bundled into probabilities is the business of the ongoing measure line (Paper 16), and no result here depends on that outcome. (ii) The full continuous phase U(1) is not claimed — what is derived is Z_4, and $Z_4 \hookrightarrow U(1)$ is an inclusion. (iii) No physical identification is made. References to standard-theory counterparts are confined to the abstract and to the form "what standard theory installs as …" in theorem statements; interpretive correspondences are quarantined with labels.

2. Choosing a time direction = choosing a complex structure

The algebraic stage of the four axes is the quaternions $\mathbb{H}$ (the {1,2,4,8} series of Paper 11).

Standard fact (mathematics): for a unit pure imaginary u, $C_u = \mathbb{R} \oplus \mathbb{R}u \subset \mathbb{H}$ is a subfield isomorphic to $\mathbb{C}$, and the polar decomposition $q = |q|e^{u\theta}$ exists — a standard fact about quaternions, not subject to verification.

> **Theorem 1 (model-specific content)**: the system's choice of time direction is realized as the choice of C_u; the modulus coincides with the norm clock of Paper 13, Theorem 5, and the argument with the phase (where the η of §4 lives) — **clock and phase are obtained simultaneously from one choice**. Complex numbers and phases are products of this choice, not axioms.
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> **Proposition 1′ (junction with the amplitude rung — proven, range-unlimited)**: for any cell $k \in \mathbb{Z}^4$, split the dressed vector $d_j = 2|k_j| + 1$ into two pairs and set the Gaussian integers $v_1 = (d_1 + id_2)/(1+i)$, $v_2 = (d_3 + id_4)/(1+i)$. Then v_1, v_2 are Gaussian integers, $N(v_1) + N(v_2) = 2s$, and each $N(v_j)$ is **odd**.
>
> **Proof** (three lines; second review R2, claude.ai): d_j odd $d_1 + d_2$ even $(1+i) \mid (d_1 + id_2)$ (divisibility).

$N(v_1)+N(v_2) = (\sum d_j^2)/2 = 2s$ (since $s = (\sum d_j^2)/4$). $d_j^2 \equiv 1 \pmod 8$ $N(v_j) = (d_a^2+d_b^2)/2 \equiv 1 \pmod 4$, hence odd. ⊠

Machine confirmation (②): 6561/6561 over $k \in \{-4..4\}^4$. The factor $(1+i)$ joins the norm rung (even $2s$) to the odd-Gaussian-norm rung.

3. The marking theorem — the all-axes-symmetric principle as a theorem

Theorem 2 (marking theorem): lattice-compatible quaternion structures (markings) number **exactly 16**, and B ⊠ (384 elements) acts on them **transitively** — each point stabilizer is the ring-automorphism group (24 elements), $384 = 16 \times 24$ (first review #1 corrected the earlier "simply transitive": simple transitivity would require $|G| = |X|$). **Enumeration (reproducible procedure)**: transport the standard quaternion product by all 384 elements of B ⊠ and deduplicate the resulting multiplication tables; 16 distinct structures remain (four with the real axis on each of the four axes; script E1). Moreover (i) the real axis runs over all four axes (E2); (ii) **of the first-order canonical readings defined per marking (real part, modulus, central character), those shared by all 16 markings (= completely B ⊠-invariant) are exactly the norm s (→ R) and ε (→ Q)** (E3; "only" is completeness with respect to this family of functions, not arbitrary cell functions — quantified per first review #2).

Consequence: which axis is time is **gauge** (chart choice); R and Q alone are viewpoint-independent invariants — the constraint surface $\Sigma = \{R^2 = s,\ Q = \varepsilon\}$ of Paper 15 is the coordinate expression of this theorem. The complete gauge data of a chart is the marking (16) plus the per-axis orientations (Z_2^4).

4. The derivation of transport phases — assignment rules vanish

When the parent's identity line persists into the daughters' cross lines across a decay (record continuity, Paper 12), the coefficient ratio

$$\eta = \frac{C_{\text{cross}}/|C_{\text{cross}}|}{C_{\text{parent}}/|C_{\text{parent}}|}$$

is fixed uniquely by the dictionary (the trigonometric identities of cos/sin) alone.

Theorem 3 (Z ⊠ quantization): on every two-stage custody path, $\eta \in Z_4 = \{1, i, -1, -i\}$ (including the 196 paths at s=9; **zero exceptions** through s=11/13; Fig. 1).

The provisional phase-assignment rules of earlier stages thereby became **unnecessary** — phase is a derived object, not a rule. $Z_4 \hookrightarrow U(1)$: the continuous phase is the inclusion target; only the inclusion is claimed.

5. The forced bit = inversion chain holonomy

Terminology note (R4): the "classes A/B" of this section are holonomy classes (relative parities of link orientations) and are **distinct** from the mirror pairs A/B (parents $\pm m$) of the measure line (Paper 16) — beware of confusion when reading both.

The diagnostic made explicit (first review #4): the z of this section is the **channel-consistent sum** (the sum of η over all custody paths within a channel) — one particular aggregate, whose measure-theoretic status (including its divergence from configuration granularity W′) is adjudicated by Paper 16. The theorems of this section are stated as **accounting properties of this aggregate** and contain no measure claim. One piece of substance behind the scope restriction (added during this review): at s=9 configuration granularity, W'_{531} becomes **sector-independent** (576=576) while W'_{333} retains dependence (40 vs 16) — the holonomy bit is not an artifact of the aggregate choice, but the "index 2" formulation below is at the channel-consistent level.

The structure of this aggregate (the exact vanishing of the fully covariant reduction, and the fact that the $|z|^2$-preserving subgroup of B ⊠ has **index 2** — Appendix A-4) demands exactly **one bit** of choice. Its identification (Fig. 3):

> **Theorem 4 (bit ≠ arrow)**: time-reversing the comparisons (full conjugation) leaves $|z|^2$ **invariant** — the forced bit is **not** the arrow of axiom 3. The link-orientation group $Z_2 \times Z_2$ (the four (o_1, o_2)) has been checked **exhaustively**: a single-link flip toggles the class, a double flip restores it (per #5: the exhaustiveness is over the orientation group; conjugation and the mirror parent are additional samples). The identity of the bit is therefore the Z_2 **chain holonomy of canonical inversion** $g \to -g$ **(complex conjugation on the dual side)**.

The system's global Z_2's are **two** (arrow; inversion holonomy), machine-verified to be independent. Inversion is the canonical structure every abelian group carries automatically — not an axiom; the inventory's "comes for free" part is where the final bit lives.

6. The canonical counting theorem — symmetrization is not a postulate

How to count identical (same-shell) daughter pairs looks at first like a choice of convention. However:

> **Theorem 5 (canonical counting)**: configuration reading (the state is the **set** of occupied cells) forces **unordered counting = symmetrization** of identical daughters. Ordered (distinguishable) counting double-counts identical physical configurations and inflates the equal-shell second-stage amplitude by a factor 2. Meanwhile, **the treatment of diagonal terms (double occupancy) and decay branches does not affect the amplitude accounting** (machine-verified 2×2 separation = ordering (unordered/ordered) × diagonal (excluded/included); same axis names as the legend of Fig. 2).

What standard theory installs as the **symmetrization postulate** (identical particles are not counted separately) appears here as a theorem of set structure. **Granularity audit** (the same audit as #4): the ordered-counting inflation acts at the level of per-configuration amplitudes z_K, so the theorem operates identically under both channel-consistent (W) and configuration-granularity (W′) aggregation — **granularity-robust**, independent of the measure line's adjudication. Note: the theorem fixes the amplitude accounting; it does not select the rule reading those values as probabilities (§1.2).

7. (Conjecture) The number ring of amplitudes

From the derived Z_4 phases and the dyadic coefficient support (powers of $\sqrt{2}$), we conjecture that **amplitudes take values in the Gaussian integers $\mathbb{Z}[i]$ (localized at $(1+i)$)**. Proposition 1′ 's $(1+i)$ junction and the ±i lemma family of Appendix A are consistent with the conjecture, but the proof is open; it is registered **as a conjecture**.

8. Honest limits

1. **No measure is selected** (§1.2 restated) — sector realization (which holonomy class is observed), aggregation rules, and the channel-specific cancellation theorems belong to the ongoing measure line and are not housed here.
2. The exhaustiveness of Theorem 3 is s=9/11/13. General-s algebraic proofs rest partly on the lemma family of Appendix A (the general proof of single-sign support remains open).
3. The identity in Theorem 5 is verified within single-shape shells — boundary cases of mixed-shape shells are unexamined.
4. Continuous phases and amplitudes have the status of inclusion and conjecture (§4, §7).
5. No physical identification is made.

Figures

- **Figure 1** (`paper14_fig1_eta_z4.png`): the transport phase η on the complex plane — all 196 paths (s=9) land exactly on Z ⊠ (531: 24/44/56/36; 333: 6/10/12/8; $\eta^2 = \pm 1$ at 80/80 and 18/18)
- **Figure 2** (`paper14_fig2_convention.png`): the 2×2 separation of canonical counting — the diagonal is irrelevant (40=40, 160=160); only ordering doubles the amplitude
- **Figure 3** (`paper14_fig3_holonomy.png`): identification of the bit — invariant under T1 (arrow reversal); single-link flips toggle A ↔ B; double flip restores

All figures are exact computations or presentations of machine-verified values (no schematics).

Appendix A: General transport-algebra lemmas (with exhaustive verification)

Lemma	Content	Verification
A-1 single-axis branch flip	a single-axis branch flip (cos ↔ sin) multiplies a nonzero transport coefficient by exactly ±i and preserves nonvanishing	exhaustive at s=9/11/13: **103,168 flips, 0 annihilations, 0 counterexamples**
A-2 single-sign support	contributing cross sums are single-signed per axis	55,267 coefficients, exhaustive
A-3 charge-like parity	$C \neq 0 \Rightarrow \varepsilon$ multiplicatively conserved (one-line parity arithmetic on the support). **As a general lemma this table is canonical; Theorem 8 of Paper 15 is its selection-rule expression**	118,944 paths + control 39.2% → 0
A-4 index 2	the $\lvert z\rvert^2$-preserving subgroup of B ⊠ has index 2 (192/384)	exhaustive

Lemma	Content	Verification
A-5 mirror flip	reflection to the mirror parent gives $\eta_B/\eta_A = -i(-1)^{\sigma_0}$ (σ_0 = axis-0 sin-branch count of the first-stage daughter pair)	m=2: 10/10; m=3: 384/384 (including coincidence of path sets)

These are algebra of transport coefficients **independent of channels and aggregation rules**. The channel-specific cancellation theorems (η ² equidistribution, A1, B2, …) concern the measure-theoretic status of channel-consistent sums and are not housed in this paper (the measure line — §8-1).

Appendix B: Verification summary (labels: ① true by construction / ② theorem + confirmation / ③ check that could have failed)

Claim	Label	Verification	Method	Script
Theorem 2, markings (enumeration, transitivity)	③	dedup of 384 transported tables	exhaustive	supplement50_marking_orbit_check.py
Theorem 2(ii), shared readings	③	invariance under all 384	exhaustive	same (E3)
Proposition 1′ , identity and oddness	**② (three-line proof)**	6561 cells	in-range confirmation	supplement49_quaternion_split_check.py
Theorem 3, Z ⊠ quantization	③	all paths s=9/11/13	exhaustive	supplement51_phase_custody_derivation.py, supplement56_s11_s13_robustness.py
Theorem 4, T1/T2	③	orientation group $Z_2 \times Z_2 = 4$ elements	exhaustive (orientation group) + samples	supplement54_bit_identification_tests.py
Theorem 4, scope note (W′ sector dependence)	③	s=9, both sectors	exhaustive	paper14_v02_wprime_check.py
Theorem 5, 2×2 separation (ordering × diagonal)	③	4 conventions × 2 channels	exhaustive	supplement58_canonical_convention.py, suppl. 60 verification
Lemmas A-1/A-2	③ (general-s proof queued)	103,168 / 55,267	exhaustive	supplement70_lemma_and_4tie.py, supplement71_bg_lemma_m3.py

Claim	Label	Verification	Method	Script
Lemma A-5	② (three-line arithmetic)	394 paths	exhaustive confirmation	supplement81_mirror_lemma_check.py

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Acknowledgments / history: verification was carried out under the two-party independent verification protocol of Claude Code (local machine verification) and claude.ai (independent re-computation). Source material: Supplements 49–54, 56–60, 70, 71, 80 (Lemma A-5: derivation claude.ai, verification Claude Code), 81 (June 12, 2026). v0.2: 5 mandatory points of the first review. v1.0: points R1–R6 of the second review — transcription of the three-line proof of Proposition 1′ (by claude.ai), label column, W′ verification script, sector terminology note, batch processing of first-round recommendations.

No physical identification is made.